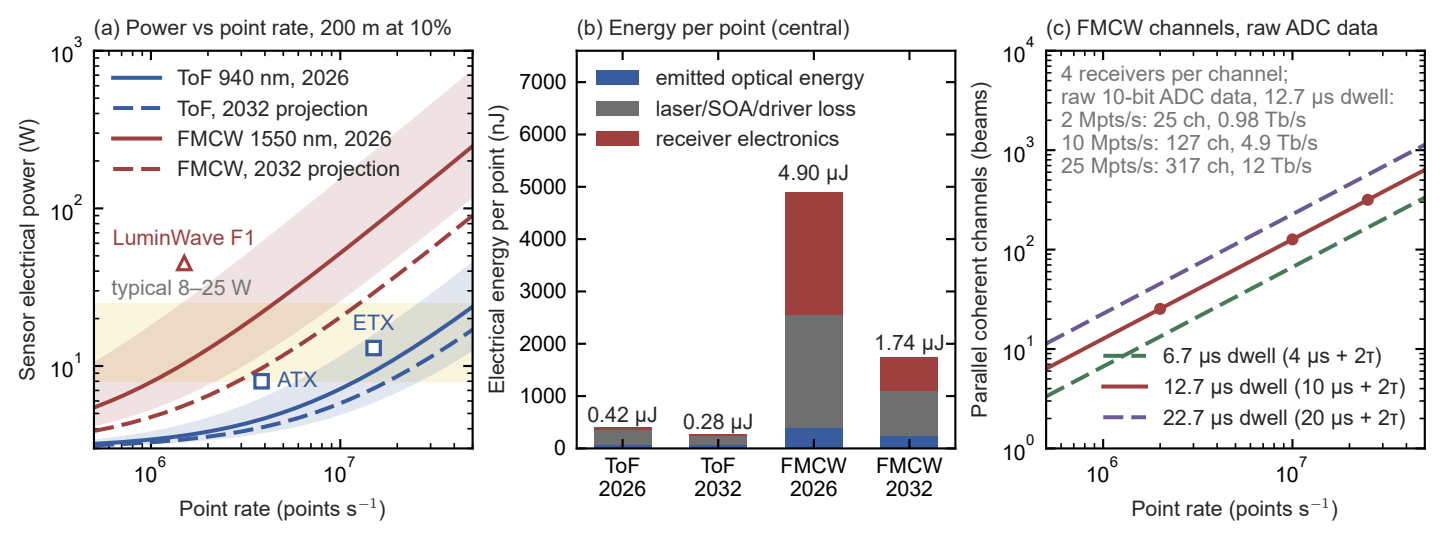
(a) Power vs point rate, 200 m at 10%
ToF 940 nm, 2026
ToF, 2032 projection
FMCW 1550 nm, 2026
FMCW, 2032 projection
LuminWave F1
typical 8–25 W
ETX
ATX
Sensor electrical power (W)
Point rate (points s^{-1})
(b) Energy per point (central)
emitted optical energy
laser/SOA/driver loss
receiver electronics
4.90 µJ
1.74 µJ
0.42 µJ
0.28 µJ
Electrical energy per point (nJ)
ToF 2026
ToF 2032
FMCW 2026
FMCW 2032
(c) FMCW channels, raw ADC data
4 receivers per channel;
raw 10-bit ADC data, 12.7 µs dwell:
2 Mpts/s: 25 ch, 0.98 Tb/s
10 Mpts/s: 127 ch, 4.9 Tb/s
25 Mpts/s: 317 ch, 12 Tb/s
6.7 µs dwell (4 µs + 2τ)
12.7 µs dwell (10 µs + 2τ)
22.7 µs dwell (20 µs + 2τ)
Parallel coherent channels (beams)
Point rate (points s^{-1})